



RIVER HIGHER INSTITUTE OF TECHNOLOGY

Diploma in Information Technology

Title: Computer Systems Architecture

Syllabus

Guided Learning Hours: 120

Level: NQF Level 6

Number of Credits: 12

Unit purpose and aim(s):

The unit covers how programs are represented and executed by modern computers, low-level machine representations of programs and data; an understanding of how computer components influence program performance, assembly level machine organization, memory system organization and architecture, functional organization, and alternative architectures. Fundamental concepts of the architectural structure and organization of computers are reviewed, including fundamental execution cycle, central processing unit, input/output unit and memory management unit, I/O controllers and processors. An analysis of the evolution of the major architectures from Complex Instruction Set Computers (CISC) to Reduced Instruction Set Computers (RISC) is carried out. The unit also addresses concepts of data encryption, authentication, hashing, and security attacks.

Von Neumann Computer Model; communication between Memory and process Unit. 5. Assembly as a low -level language and why it lacks high-level conveniences such as variables and functions. 6. The difference between CISC (Complex Instruction Set computer) and RISC (Reduced Instruction Set Computer).	4.2 Define bus characteristics. 4.3 Describe general and special-purpose registers. 4.4 Identifying memory operations and the relationship between memory address register, memory data register and program counter. 4.5 Describe memory capacity. 4.6 Define Point-to-point vs multipoint. 4.7 Describe the motherboard layout. 4.8 Describe the instruction set format. 5.1 Explain low-level language basics. 5.2 Describe how assembler reads a sample program, and converts each line of code into one CPU-level instruction. 5.3 Describe how assembly language is compiled. 5.4 Define assembly language instruction format. 6.1 Describe the CISC architecture. 6.2 Describe the limitations of CISC architecture. 6.3 Describe RISC features. 6.4 Describe Very Long Instruction Word (VLIW) architecture. 6.5 Describe EPIC (Explicitly Parallel Instruction Computer) architecture. 6.6 Differentiate logical vs physical addresses. 6.7 Define cache memory, describe the difference between cache and virtual memory.
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7. Understanding how the processing speed or program execution is determined primarily by the ability of Input/Output (I/O) operations to stay ahead of the processor. 8. Computer peripherals, their classifications, how they are connected, the characteristics and features of Real-Time systems. 9. The difference in layout between PC, mini, and mainframe systems, including mass storage systems.	7.1 Describe I/O speed and coordination issues. 7.2 Describe I/O device interface issues. 7.3 Describe Input/Output module functions. 7.4 Define the CPU interrupts. Explain the use of interrupts. 7.5 Define Direct Memory Access (DMA). 7.6 Define data bus configuration architecture. 7.7 Describe different external bus and port interfaces. 7.8 Explore the structure of an operating system's I/O subsystem. 7.9 Discuss the principles of I/O hardware and its complexity. 7.10 Provide details of the performance aspects of I/O hardware and software. 8.1 Describe storage devices and their data access time. 8.2 Describe the hard disk layout format. 8.3 Describe the CD-ROM layout. 8.4 Distinguish between Hard and Soft Real-Time systems. 9.1 Differentiate PC and mainframe components. 9.2 Describe Symmetric Multiprocessing System (SMP) 9.3 Describe client-server architecture. 9.4 Define parallel computing. 9.5 Describe the physical structure of secondary and tertiary storage devices
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10. Overview of operating systems concept, architecture, structure and the fundamental model. 11. How to protect the system resources and the external environment of a system.	and the resulting effects on the uses of the devices. 9.6 Explain the performance characteristics of mass-storage devices. 9.7 Discuss operating -system services provided for mass storage, including RAID and HSM. 10.1 Describe the services provided by an operating system to users, processes and other systems. 10.2 Describe the relationship between hardware and the operating system. 10.3 Explain how operating systems are installed and customized and how they boot. 11.1 Discuss security threats and attacks. 11.2 Explain the fundamentals of encryption, authentication and hashing. 11.3 Examine the uses of cryptography in computer. 11.4 Describe the various countermeasures to security attacks.
Methods of evaluation: A 3 hours examination constituting of section A essay questions with 40 marks and section B with 5 essay questions (each carrying 20 marks), and students will only choose 3 out of the 5. The whole paper will be out of 100%, and will contribute 70% to the final mark. Candidates also undertake coursework of assignments and tests which contributes 30% to the final mark.	

Recommended Learning Resources: Computer Systems Architecture

Text Books	• Introduction to Computing Systems; From Bits and Gates to C and beyond 2nd Edition. • Computer Organization and Design Fundamentals by David Tarnoff. • Principles of Computer Architecture Miles Murdocca and Vincent Heuring. • Computer Organization and Architecture Designing for Performance Eighth Edition by William Stallings
Study Manual	MSC Study pack
Websites	
Software	